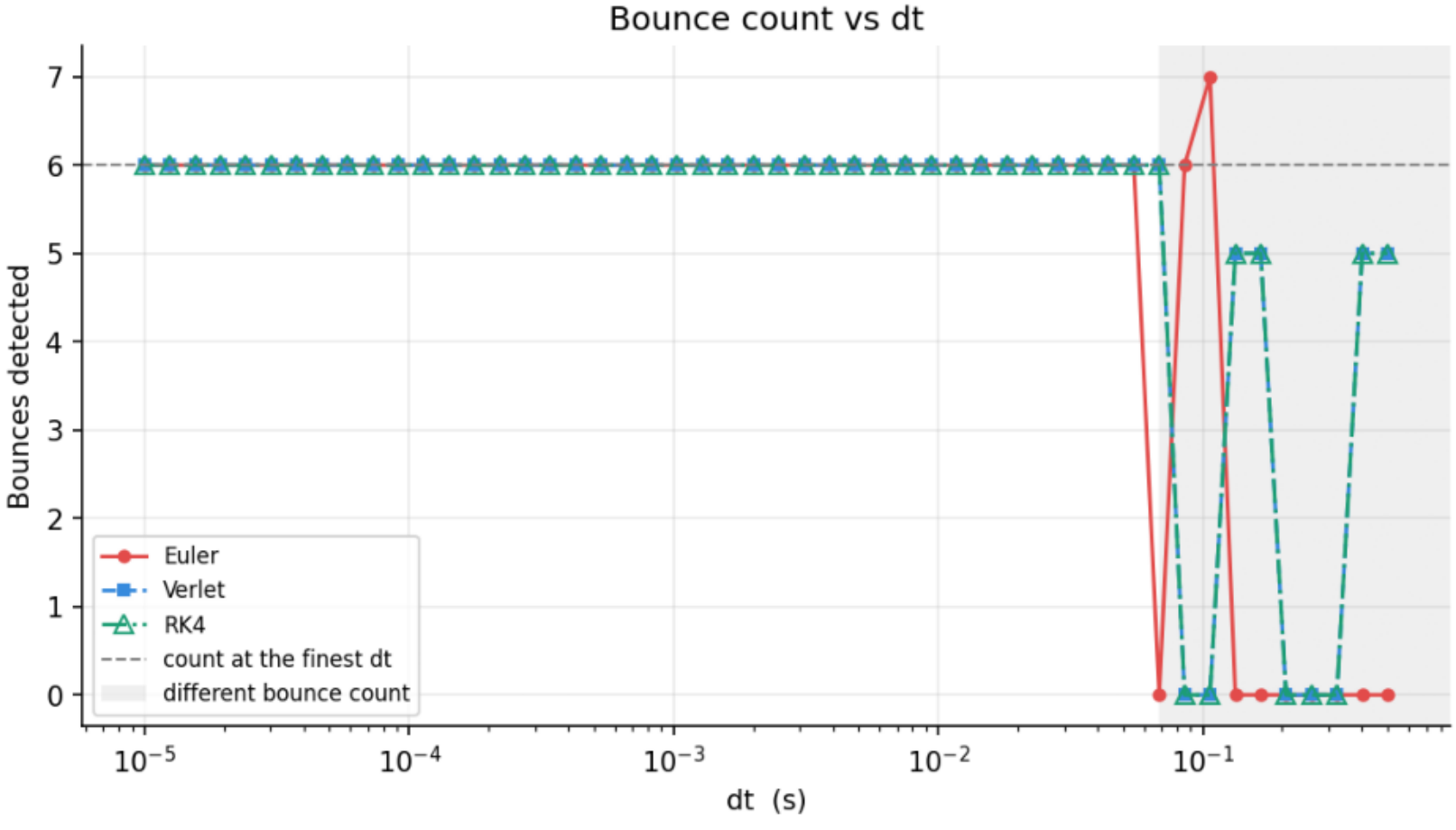
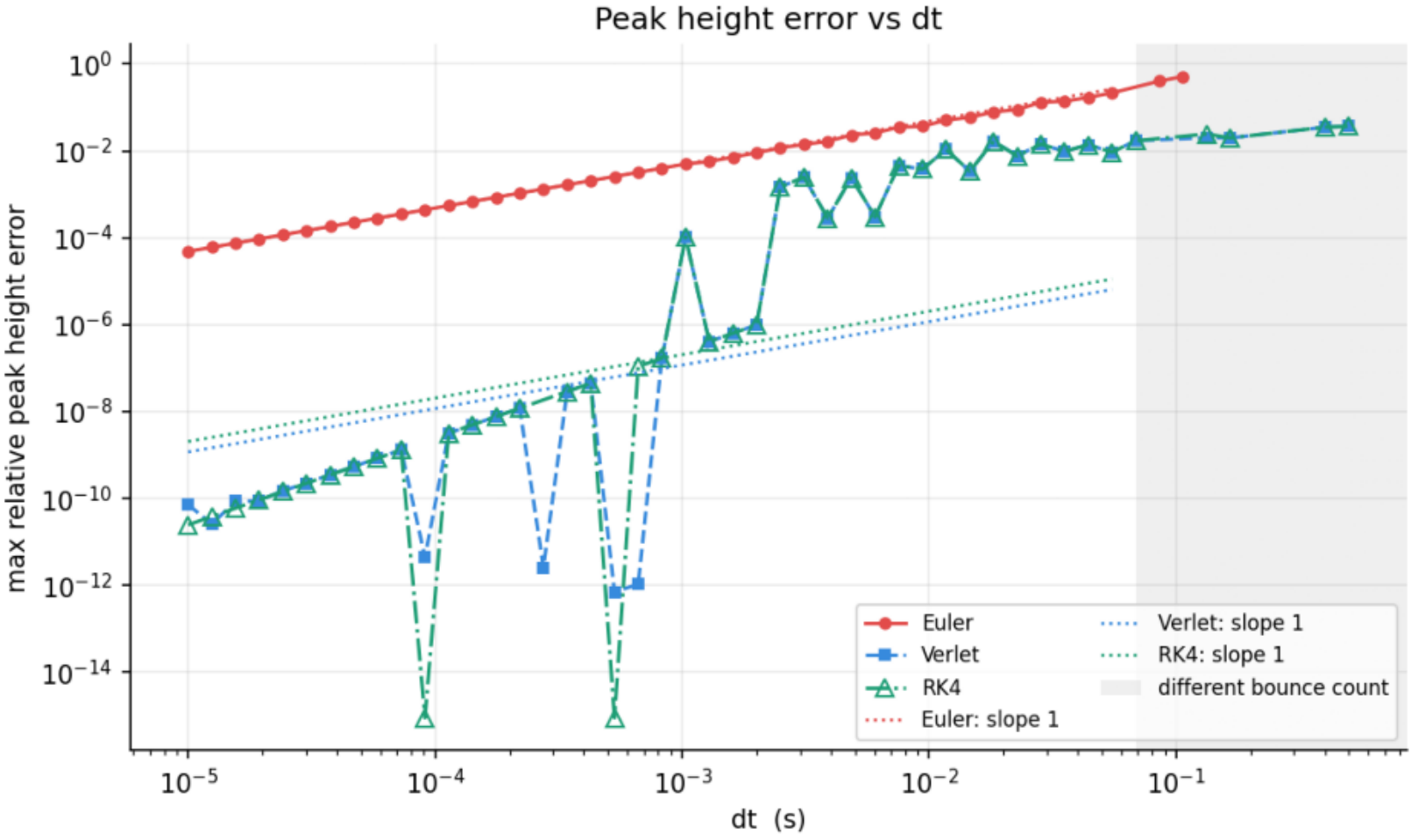
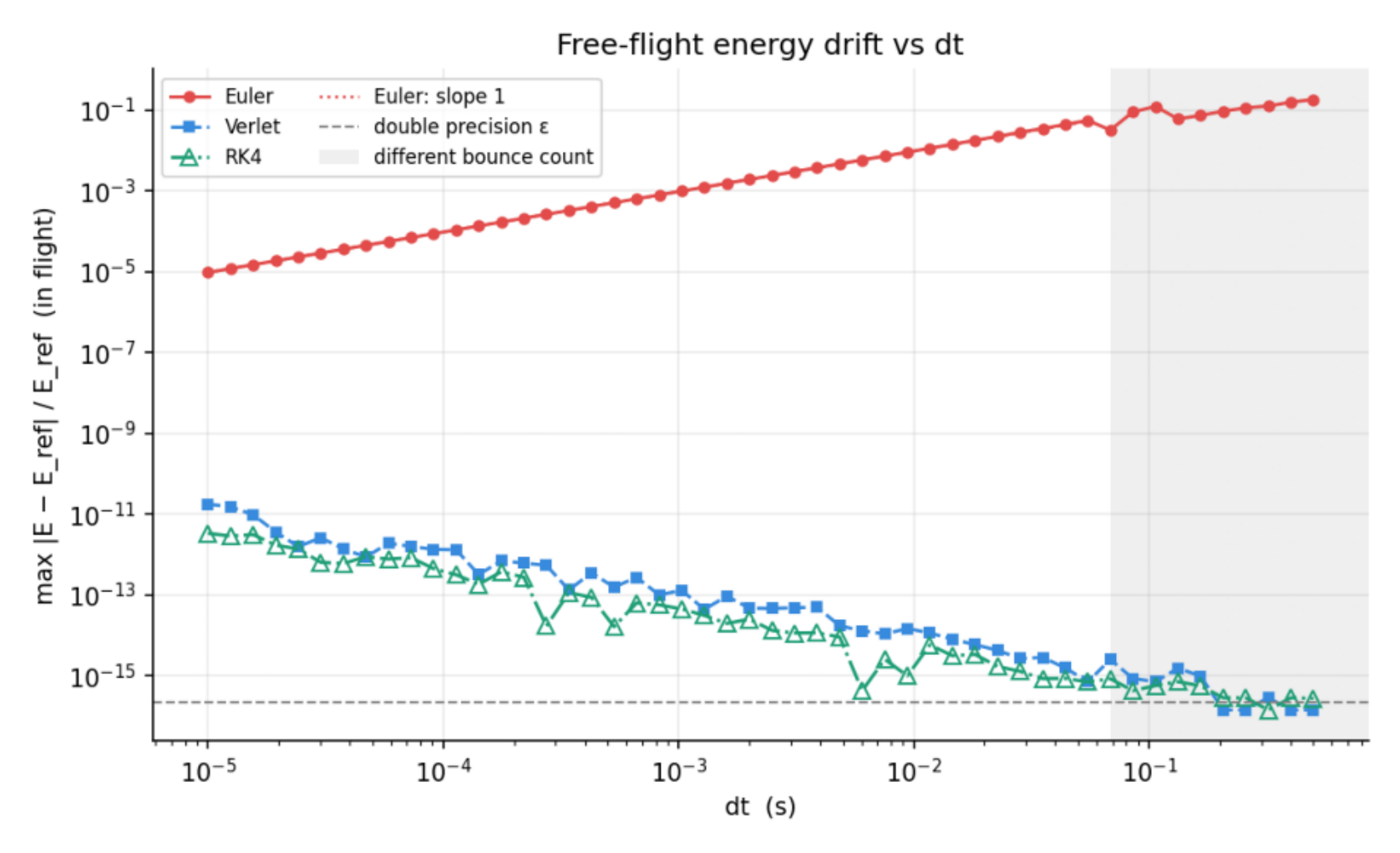
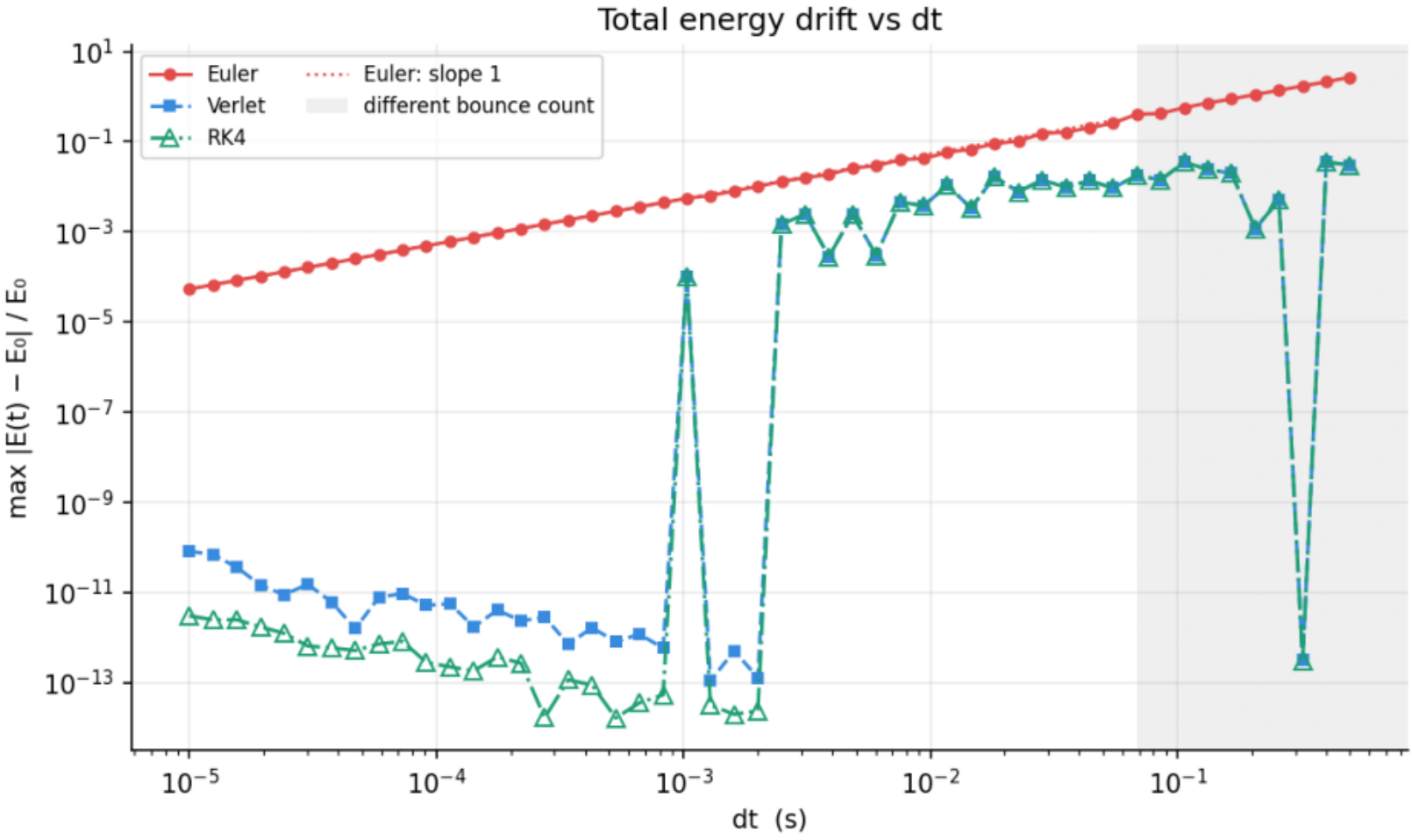
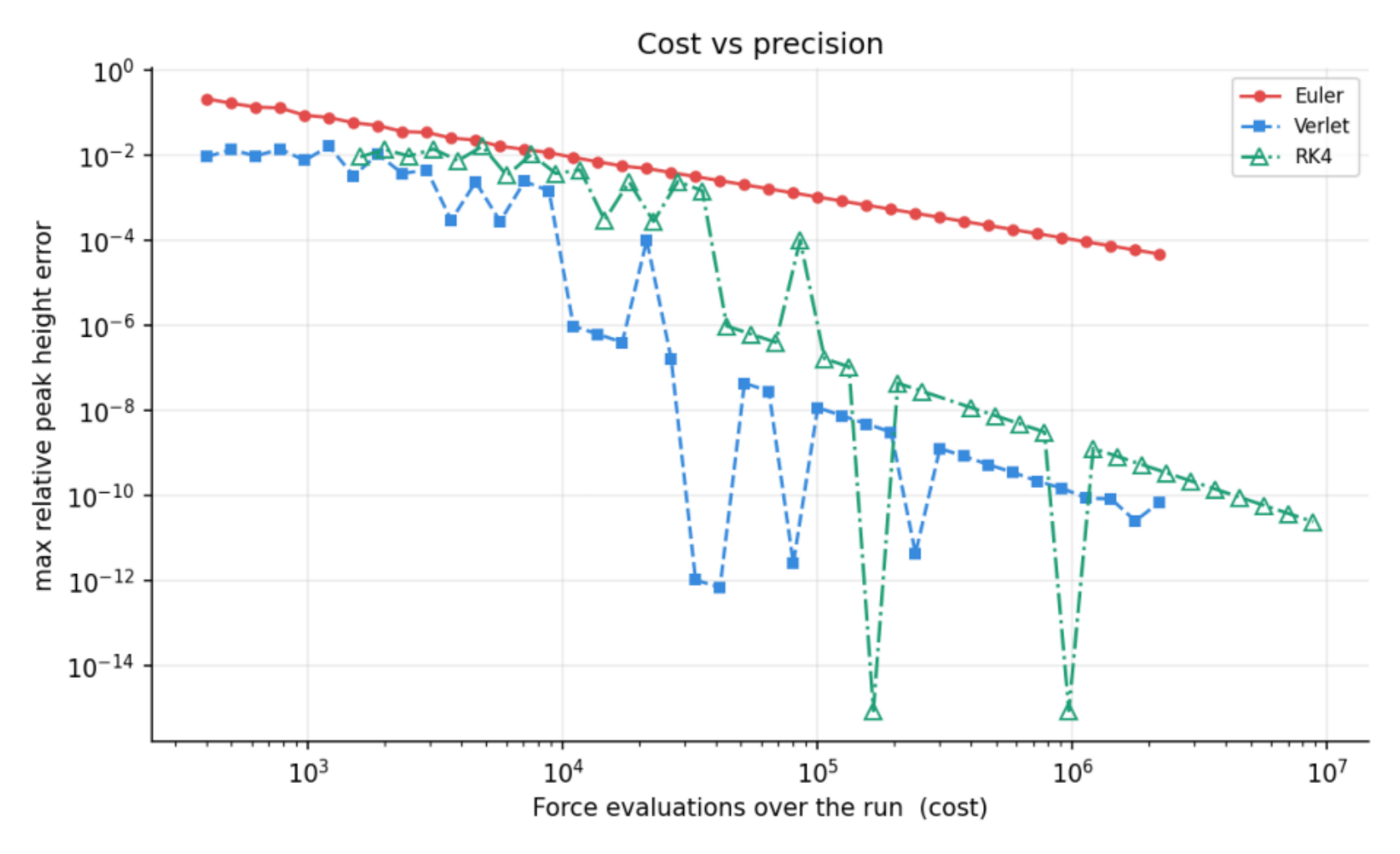
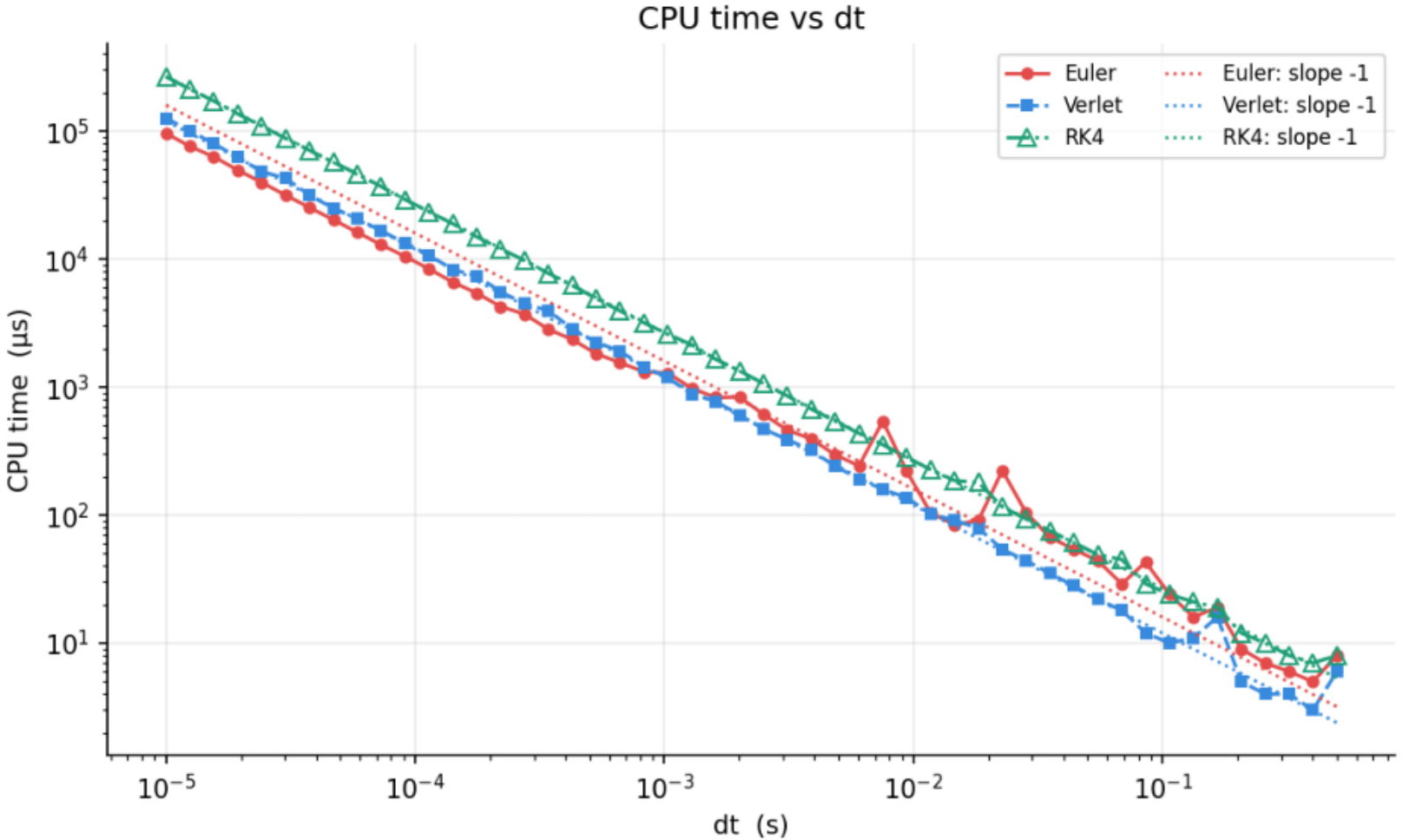

Bounce count vs dt
Bounces detected
dt (s)
Euler
Verlet
RK4
count at the finest dt
different bounce count

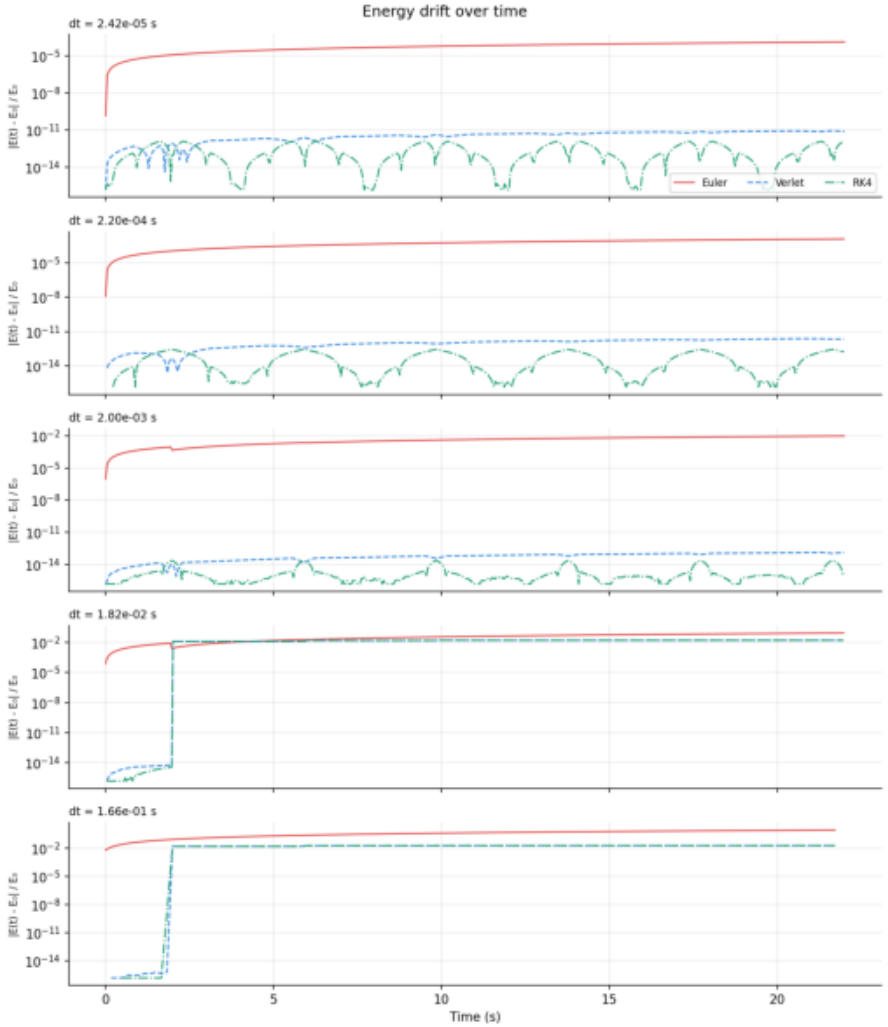

Peak height error vs dt
max relative peak height error
dt (s)
Euler
Verlet
RK4
Euler: slope 1
Verlet: slope 1
RK4: slope 1
different bounce count


Free-flight energy drift vs dt
Euler
Verlet
RK4
Euler: slope 1
double precision ε
different bounce count
max |E − E_ref| / E_ref (in flight)
dt (s)
10^{-1}
10^{-3}
10^{-5}
10^{-7}
10^{-9}
10^{-11}
10^{-13}
10^{-15}
10^{-5}
10^{-4}
10^{-3}
10^{-2}
10^{-1}


Total energy drift vs dt
Euler
Verlet
RK4
Euler: slope 1
different bounce count
max |E(t) − E₀| / E₀
dt (s)
10^1
10^-1
10^-3
10^-5
10^-7
10^-9
10^-11
10^-13
10^-5
10^-4
10^-3
10^-2
10^-1


Cost vs precision
Euler
Verlet
RK4
max relative peak height error
Force evaluations over the run (cost)
10^{0}
10^{-2}
10^{-4}
10^{-6}
10^{-8}
10^{-10}
10^{-12}
10^{-14}
10^{3}
10^{4}
10^{5}
10^{6}
10^{7}


CPU time vs dt
Euler
Verlet
RK4
Euler: slope -1
Verlet: slope -1
RK4: slope -1
CPU time (µs)
dt (s)
10^{5}
10^{4}
10^{3}
10^{2}
10^{1}
10^{-5}
10^{-4}
10^{-3}
10^{-2}
10^{-1}


Energy drift over time
dt = 2.42e-05 s
dt = 2.20e-04 s
dt = 2.00e-03 s
dt = 1.82e-02 s
dt = 1.66e-01 s
|E(t) - E₀| / E₀
Euler
Verlet
RK4
Time (s)
0
5
10
15
20